

- What is the difference between an AI engineer and an ML engineer? If a company asked you to train a custom model from scratch, is that your job? Why or why not?

Answer:

An AI Engineer is the professional who builds software systems around the models while an ML Engineer is the professional who trains the models. No it is not my job, my role is to build software, customer-facing solutions with these model's API.

- A language model is described as a “text prediction machine.” What does that mean concretely? What is it predicting, and how does it choose?

Answer:

A language model is trained on a massive amount of words to the extent that it becomes so good at predicting the next word. How it actually predicts is through mathematics, cosine similarity to bring the next word through vector similarity, this allows us to not use keywords but instead an array of 1536 numbers that is similar to another set of words with a highly similar array of numbers. For instance, similarity of 1.0 of almost that will definitely be predicted compared to words with similarity of 0 or close to 0, that will not be predicted by the language models.

- Explain the full RAG pipeline in five sentences. Start from the user asking a question and end with the user seeing an answer with citations.

Answer:

1. The user asks a question
2. The question gets transformed to numbers
3. It is sent to the vector database to compare and see if there is anything that can be retrieved from the document.

4. The retrieved information is added to the context prompt and sent to the models API.
5. A more personalised response is sent back to the user.

- Why does the model not know what is in a user's uploaded document? If a user uploads a PDF to your system, what are the exact steps needed before the model can answer questions about it?

Answer:

You will need to use rag to help the model know what is in your document. The model is trained on billions and billions of parameters, not your company's business data, instead of fine tuning, simply use rag to make life easier.

- Your capstone uses temperature=0.1 for RAG answers. A product manager asks you to change it to 0.7 because "the answers sound too robotic." What is the risk? What do you tell them?

Answer:

The higher the temperature, the more the randomness, it is more suitable for creative work or where creativity is a feature, I would also recommend that we try out temperature = 0.25, and look at the feedback from the model and product manager.

- What is the difference between Top-K and Top-P? Describe a specific probability distribution where Top-K=50 selects tokens that Top-P=0.9 would exclude.

Answer:

Top-K is for number of tokens that a model can choose from it is a filter however top-p at 0.9 is just as good because top - p is a smarter filter.

- An LLM has a 128,000 token context window. A user uploads a 200-page document. Can you send the entire document in one API call? Why or why not? What do you do instead?

Answer:

Why would i send the entire document in one api call tho?, each call costs money and as an ai engineer, i would be trying to find the shortest route to reduce cost and get the best response. For this, i would get the most relevant context from the document, and include it to the context window, this way, the model is more aware of what is user is asking for

- What is the difference between prompt engineering and context engineering? Give an example where the prompt is perfect but the context makes the answer wrong.

Instructions to the model versus everything you put in the context window. I have no example at the moment.

- Embeddings represent text as lists of numbers. What does “similar meaning” mean in terms of these numbers? How do you measure it? What is the formula called?

Answer:

Similar meaning is what we say when we convert text to numbers and see that the 1536 numbers for token a is 0.96 similar to the 1536 numbers for token b hence it gets predicted. It is measured in the vector database. The formula is called cosine similarity.

- If inference is just sending an HTTP request to an API, why is an entire 6-week bootcamp needed to learn AI engineering? What is the hard part that is not the API call itself?

Answer:

It is the Retrieval Augmented Generation, the model cannot determine that from the api call without it being implemented by the ai engineer. It is to

prevent hallucinations and provide guardrails and maintain cost and improve latency that requires a 6-week bootcamp to be important.

- Why would you use different temperature values for different features in the same application? Give three examples: one that needs temperature=0.1, one that needs 0.3, and one that needs 0.7. Explain why each value is appropriate.

Answer:

It all depends on the feature

0.1 - when you need determinable, repetitive responses

0.3 - Factual consistent answers but not degenerative answers

0.7 - When variety or brainstorming is valuable.